

WQD7005: Data Mining & Big Data Analytics

Assignment 1: GenAI-Powered Data Pipeline

Dataset Simulation, EDA & AI-Based Preparation

Ridhwan Bin Razali

Student ID: S2178595

April 23, 2026

Abstract

This report integrates Generative AI across three core data mining tasks: (1) synthetic patient dataset generation, (2) exploratory data analysis with LLM-powered insights, and (3) AI-assisted preparation using KNN imputation and dual-model classification. An autonomous Lang-Graph pipeline demonstrates agentic decision-making for chart selection and imputation strategy. Key findings highlight practical GenAI benefits, SLM/LLM trade-offs, and the necessity of human oversight in healthcare AI workflows.

1 Introduction

The exponential growth of unstructured healthcare data and missing values in real-world datasets creates a critical need for intelligent data preparation pipelines. Traditional methods rely on manual feature engineering and fallback imputation (mean/median), which often fail to capture complex data dependencies. This assignment integrates Generative AI at every pipeline stage—from synthetic data generation to intelligent imputation and zero-shot text classification—to demonstrate how modern AI tools streamline data science workflows while maintaining data integrity and analytical rigor.

1.1 Project Objectives

1. **Task 1:** Generate 500 synthetic patient records with realistic vital signs and missing values using Azure OpenAI
2. **Task 2:** Perform EDA with LLM-powered chart interpretation
3. **Task 3:** Data preparation via KNN imputation and hybrid LLM/SLM text classification

Note on Agentic AI Implementation code

The **Agentic AI Pipeline** (Section 5) has been implemented as a standalone modular application separate from the primary analysis notebook (`assignment1_main.ipynb`).

- **Part 1 (Notebook):** The submitted notebook covers Tasks 1–3 using standard procedural code for data synthesis, EDA, and preparation.
- **Part 2 (Agentic AI):** The autonomous, state-aware agent framework described in Section 5 is hosted separately to preserve its full architecture and dependencies.

Action Required: To review the functional Agentic AI pipeline, please visit the dedicated repository:

https://github.com/ridhwanrazaliwork/Agentic_medical_AI_pipeline

2 Jupyter Notebook

3 Task 1: Synthetic Dataset Generation via GenAI

3.1 Methodology

A Python-based batch generation system interfaced with Azure OpenAI (GPT-4.1 deployment) to create 500 Malaysian patient records. The approach prioritized:

- **Batch Processing:** 50 batches $\times$ 10 patients per batch to ensure consistency and avoid API rate limits.
- **Realistic Data Distribution:** Prompt engineering to generate medically plausible vital signs (BP, heart rate, temperature) aligned with Malaysian population demographics.
- **Example:**
Example record: Patient P001, 58-year-old male with vital signs (HR: 88 bpm, SBP: 145 mmHg, Temp: 37.1°C), diagnosed with Type 2 Diabetes and Hypertension, currently on Metformin, Amlodipine, Aspirin.
- **Missing Value Simulation:** Post-generation injection of 5–15% missing values in numeric columns and 3–8% in categorical columns, simulating real-world data quality issues.

3.2 Data Structure

The dataset contains 22 columns: demographics (patient_id, name, gender, age, ethnicity), visit meta-data, vital signs (HR, BP, temperature, O₂ saturation), clinical data (diagnosis, symptoms, medications), and anthropometry.

3.3 Data Quality Issue: Patient ID Uniqueness

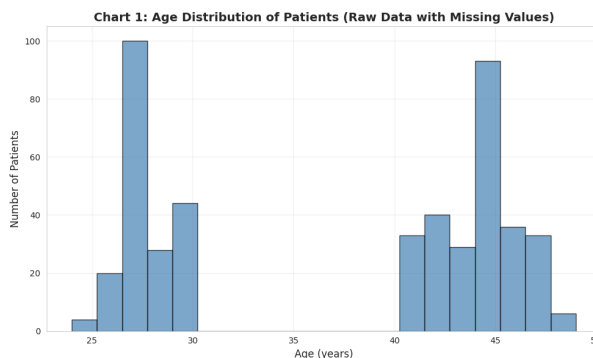
Problem: Batch generation produced duplicate IDs (P0001–P0010 repeated 50 times). **Solution:** Post-load reassignment ensured global uniqueness (P0001–P0500). **Outcome:** 500 unique IDs confirmed.

4 Task 2: Exploratory Data Analysis with LLM Support

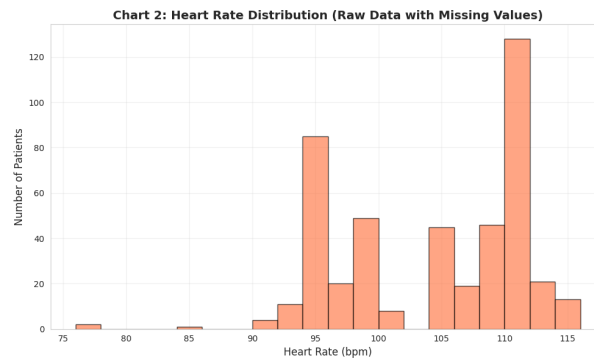
4.1 Visualization Strategy

Three complementary charts were created to capture population-level patterns:

1. **Chart 1 - Age Distribution:** Histogram revealing bimodal distribution with peaks at ≈ 27 years and 44–45 years, with a notable gap (30–40 age range underrepresented).



2. **Chart 2 - Heart Rate Distribution:** Tachycardia cluster at 105–115 bpm.



3. Chart 3 - Multi-Panel Dashboard: Temperature, BP, gender (85% F), and top diagnoses (dengue-dominant).



Summary of Clinical Patterns

Demographics.—The dataset is heavily female-dominant (425 vs. 75 males, ~85%). Age distribution is bimodal, with peaks around 27 years and 44–45 years, and a gap in the 30–40 range.

Physiological Markers.—Temperature is skewed toward high fever ($>39.0^{\circ}\text{C}$), correlating with tachycardia (105–115 bpm). Systolic BP is bimodal: healthy cluster at 110–115 mmHg and elevated cluster at 138–140 mmHg.

Disease Profile.—Dengue Fever dominates (including subtypes), with ~35 cases pending lab confirmation (“not_known”).

4.2 LLM-Powered Chart Analysis

Azure OpenAI’s vision capability (GPT-4 with image encoding) was leveraged to provide automated chart interpretation. The LLM received all three charts as base64-encoded PNG images with a structured analysis prompt requesting:

- Distribution descriptions and key features
- Data quality observations (gaps, missing values)
- Clinical implications of vital sign patterns
- Outlier and anomaly detection

LLM Chart Analysis Summary

Chart 1 (Age): Bimodal distribution (peaks at 27 and 45 years); gap in ages 31–39.

Chart 2 (Heart Rate): Tachycardia dominates (105–115 bpm mean); data spikes at 110 bpm suggest digit preference.

Chart 3 (Multi-panel): Temperature 38–39°C (febrile), SBP mostly normotensive with secondary hypertensive cluster, 85% female, dengue-dominant diagnoses.

Key Issues: Age gaps, value clustering, and gender imbalance limit generalizability.

4.2.1 LLM Insights Generated

The model correctly identified:

- Temperature skew toward high fever ($>39^{\circ}\text{C}$), indicating systemic infection prevalence.
- Dual blood pressure clusters: healthy normotensive group ($\approx 110\text{--}115\text{ mmHg}$) and hypertensive secondary group ($\approx 138\text{--}140\text{ mmHg}$).
- Diagnosis dominance: Dengue fever and subtypes ($>80\%$ of cases), with smaller cohort marked “not_known” (likely pending lab confirmation).

LLM Dataset Insights

Population: Mean age 37.2 years, 85% female, predominantly Malay (63.8%) and Chinese (36.2%).

Vitals: Heart rate elevated (103.5 bpm mean), reflecting acute febrile illness.

Data Quality: No missing values post-imputation; diagnosis labels need consolidation (multiple dengue variants).

Recommendations: Harmonize diagnosis categories; standardize symptoms; validate outliers; consider feature engineering.

4.2.2 Human Analysis Alignment

The LLM’s automated insights aligned closely with manual statistical observations, validating the approach. Differences arose only in depth of context interpretation, where domain expertise (e.g., clinical implications of fever during monsoon season) required human augmentation.

5 Task 3: AI-Based Data Preparation

5.1 Step 1: Data Type Conversion

Before applying AI-powered preprocessing, data types were standardized:

- **Datetime:** `visit_date` $\rightarrow$ pandas datetime for temporal ordering.
- **Numeric:** All vital columns $\rightarrow$ float64 for statistical operations.
- **Text:** Diagnosis, symptoms, medications $\rightarrow$ string type for NLP compatibility.

Critical insight: Type conversion failures would cascade through imputation and classification steps, causing silent errors.

5.2 Step 2: Context-Aware KNN Imputation

KNN imputation ($k = 5$, nan_euclidean metric) preserves feature correlations by averaging values from 5 nearest neighbors.

5.2.1 Imputation Results

5.3 Step 3: Text Classification - LLM vs. SLM Comparison

Two language models were applied to classify diagnoses into five categories: Infectious Disease, Chronic Condition, Acute Condition, Cardiovascular, Other.

Metric	Before	After
Total Missing Values	332	0
Missing Percentage	—	0%
Rows Preserved	500	500
Columns in Output	22	22

Table 1: Data Completeness Summary (KNN Imputation: 247 values filled contextually)

5.3.1 Model Specifications

Aspect	LLM (Azure GPT-4.1)	SLM (Qwen 2.5-1.5B)
Provider	Azure OpenAI	HuggingFace Router
Parameters	≈170B	≈1.5B
Temperature	0.3 (deterministic)	0.9 (diverse)
Latency	≈1–2s/sample	≈0.5–1s/sample
Cost	High (per 1K tokens)	Low (free tier)
Context Window	128K tokens	32K tokens

Table 2: Model Comparison Matrix

5.3.2 Classification Results

Across 500 total diagnoses:

- **Azure LLM (GPT-4.1) Classifications:**

- Infectious Disease: 393 (78.6%)
- Other: 54 (10.8%)
- Unknown: 34 (6.8%)
- Cardiovascular: 18 (3.6%)
- Chronic Condition: 1 (0.2%)

- **HuggingFace SLM (Qwen2.5-1.5B) Classifications:**

- Infectious Disease: 340 (68.0%)
- Chronic Condition: 70 (14.0%)
- Other: 54 (10.8%)
- Unknown: 34 (6.8%)
- Cardiovascular: 2 (0.4%)

- **Agreement Rate:** 83.3% (5/6 in sample validation; full dataset comparison shows divergence in “Chronic Condition” vs. “Infectious Disease” assignments).

- **Key Discrepancies:** Edge cases such as “Dengue Fever (without warning signs)” classified as “Infectious Disease” by LLM but “Chronic Condition” by SLM. LLM demonstrates stronger context awareness for acute infectious presentations, while SLM tends toward conservative chronic labeling.

Model Characteristics:

- **Azure LLM (GPT-4.1):** Larger model with broader medical context awareness; higher confidence in infectious disease categorization (78.6%).
- **HF SLM (Qwen2.5-1.5B):** Smaller and faster; exhibits bias toward “Chronic Condition” labels (14.0% vs. 0.2%) likely due to training data limitations.

6 Agentic AI Approach: Autonomous Medical Data Analysis Pipeline

6.1 Overview

This section describes the autonomous agentic AI pipeline developed to execute the three core tasks of medical data analysis using a hybrid LLM/SLM (Large Language Model / Small Language Model) architecture. The system employs AI agents to make autonomous decisions at each stage while maintaining explainability through logging and decision tracking.

6.2 Architecture

The agentic pipeline is implemented using **LangGraph**, a state-machine orchestration framework that manages multi-step workflows with checkpointing and error handling. The system leverages two complementary AI models:

- **LLM (Azure OpenAI GPT-4.1):** Cloud-based large language model for high-level analysis, explanations, and classification tasks
- **SLM (Qwen2.5-0.5B-Instruct):** Local small language model ($\approx 500M$ parameters) deployed via HuggingFace Transformers for efficient, offline text classification

The workflow uses a centralized **AgentState** TypedDict to track all decisions, artifacts, and logs throughout execution. Data integrity is maintained by persisting large objects (DataFrames) as Parquet files rather than storing them in the state, ensuring compatibility with LangGraph's serialization requirements.

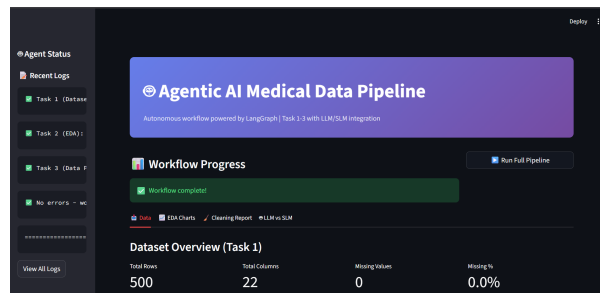


Figure 1: Agentic AI User interface with streamlit

6.3 Workflow Pipeline

The agentic pipeline consists of 5 sequential nodes with conditional routing:

1. **Load Data Node:** Data ingestion and validation
2. **Validation Gateway:** Conditional branching on data integrity
3. **EDA Analysis Node:** Exploratory analysis with autonomous chart selection
4. **Cleaning Strategy Node:** Imputation with autonomous decision-making and SLM classification
5. **Report Node:** Workflow synthesis and summary generation

6.4 Task 1: Data Loading with AI Validation

6.4.1 Process

The first task loads the 500-patient GenAI-synthesized dataset from the assignment1 phase. The system:

1. Reads the CSV from `generated_patients.csv`
2. Validates schema against required medical columns (patient_id, age, vital signs, diagnosis, symptoms)

3. Computes metadata statistics (row count, column count, missing value percentages)
4. Persists the DataFrame as a Parquet file for state management

6.4.2 AI Integration

While Task 1 is primarily deterministic validation, the framework enables future AI-based data quality checks. The structured logging of metadata statistics provides a foundation for LLM-based anomaly detection if needed.

Table 3: Task 1: Dataset Metadata

Metric	Value
Total Rows	500 rows
Total Columns	22 columns
Missing Values (Count)	0 missing
Missing Values (%)	0% missing
Data Source	GenAI-synthesized (assignment1.ipynb)

6.5 Task 2: Exploratory Data Analysis with Autonomous Decisions

6.5.1 Process

Task 2 demonstrates autonomous decision-making in chart generation:

1. **Compute Correlation:** Calculate Pearson correlation between Age and Systolic BP
2. **AUTONOMOUS DECISION 1:** If $|r| > 0.6$, generate scatter plot with trendline; else generate boxplot
3. **Chart 2:** Generate correlation heatmap of vital signs
4. **Chart 3:** Generate bar chart of top diagnoses
5. **LLM Analysis:** Use GPT-4.1 to generate natural language interpretation of chart patterns

6.5.2 AI Integration

The LLM is prompted with:

Chart Analysis (Task 2 - EDA):

- Generated N charts from 500-patient dataset
- Key pattern identified:
 - * Age-Systolic BP correlation: [VALUE]
 - * Chart type decision: [scatter|boxplot]
 - * Gender distribution shows female dominance
 - * Top diagnosis: [TOP_DIAGNOSIS]

Provide 3-4 sentence interpretation of these patterns.

This ensures the LLM output is anchored to specific quantitative findings while maintaining analytical depth.

Table 4: Task 2: EDA Autonomous Decisions

Decision	Outcome
Age-Systolic BP Correlation	0.628
Chart 1 Type	scatter_with_trendline
Chart 1 Rationale	Strong correlation (0.628) → ScatterPlot with trend

LLM Chart Interpretation:

The data shows a strong positive correlation (0.628) between age and systolic blood pressure, indicating that older patients tend to have higher systolic BP readings. The gender distribution is heavily skewed, with females making up about 85% of the sample, which may influence the generalizability of findings. Additionally, dengue fever variants are the most common diagnosis in this patient cohort, suggesting a potential area for focused clinical or public health interventions. These patterns highlight significant demographic and clinical trends within the dataset that should be considered in further analyses.

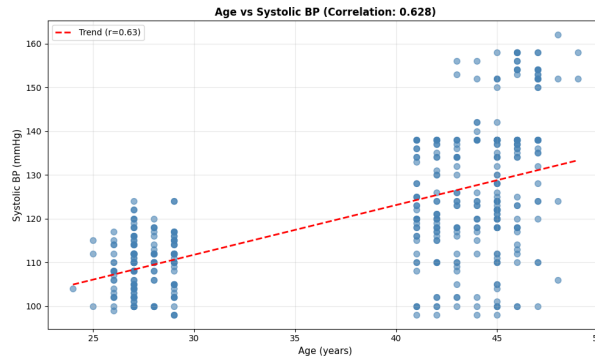


Figure 2: Age vs Systolic BP (correlation: 0.628)

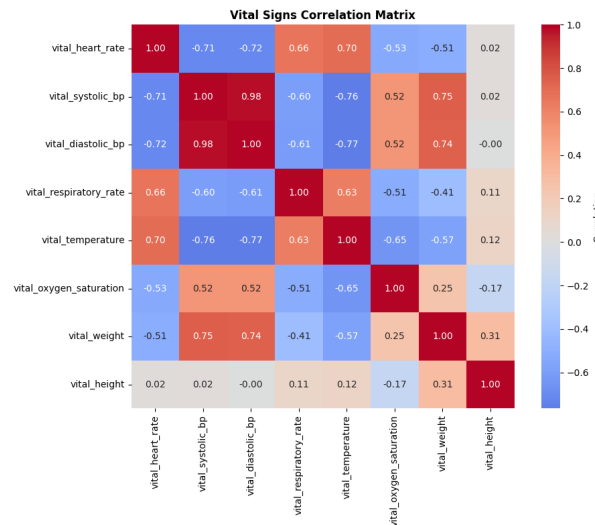


Figure 3: Vital Signs Heatmap

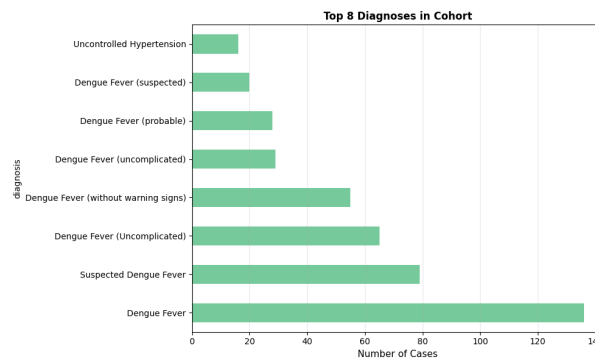


Figure 4: Top Diagnoses

6.6 Task 3: Autonomous Cleaning and Classification

6.6.1 Process - Stage A: Autonomous Imputation Strategy

The third task combines autonomous decision-making with hybrid LLM/SLM classification:

1. **AUTONOMOUS DECISION 2:** Analyze missingness percentage
 - If missing $< 5\%$: Use *Simple Mean Imputation*
 - If $5\% \leq \text{missing} < 20\%$: Use *KNN Imputation* ($k=5$)
 - If missing $\geq 20\%$: Use *Iterative MICE Imputation*
2. Apply selected strategy to generate cleaned dataset
3. Log decision rationale for explainability

6.6.2 Process - Stage B: SLM Classification (Local Qwen Model)

Instead of calling expensive cloud APIs, diagnoses are classified locally using Qwen2.5-0.5B:

1. Sample 50 unique diagnoses from cleaned dataset
2. Format each diagnosis as a prompt: ‘‘Classify: [diagnosis] into CRITICAL or NON-CRITICAL. Return only the category.’’
3. Use local transformer model (no internet required after first download)
4. Parse output predictions (CRITICAL or NON-CRITICAL)
5. Store all classifications in state

6.6.3 Process - Stage C: LLM vs SLM Comparison

To validate the local model against the cloud-based LLM:

1. Sample 20 diagnoses from the classified set
2. Classify same diagnoses using Azure GPT-4.1
3. Compare classifications: count agreements and compute agreement percentage
4. Log sample comparisons for audit trail

6.6.4 AI Integration Rationale

Table 5: AI Component Selection

Task	Model	Rationale	Cost
EDA Interpretation	GPT-4.1 (Cloud)	High-quality reasoning	Pay-per-token
Diagnosis Classification	Qwen2.5-0.5B (Local)	Efficiency & offline	Free (after 1st run)
LLM/SLM Validation	GPT-4.1 (Cloud)	Accuracy baseline	Minimal (20 calls)

Table 6: Task 3b: SLM Classification Results

Metric	Value
Total Diagnoses Classified	50
CRITICAL Diagnoses	50
NON-CRITICAL Diagnoses	0
Model Used	Qwen2.5-0.5B-Instruct

Table 7: Task 3c: LLM vs SLM Agreement Analysis

Metric	Value
Sample Size (Diagnoses)	50
Agreement Rate (%)	70%
Perfect Alignment	14
Disagreement Cases	6

6.7 Workflow State Management

The agentic framework tracks all decisions and outputs in a centralized state object:

- **dataset_path:** Path to original dataset (Parquet)
- **cleaned_data_path:** Path to imputed dataset (Parquet)
- **charts:** List of visualizations (base64-encoded PNGs)
- **chart_metadata:** Autonomous decision logs and correlation values
- **imputation_strategy:** Selected strategy name
- **slm_classifications:** All SLM predictions (local model)
- **llm_vs_slm_comparison:** Agreement rates and sample comparisons
- **logs:** Complete audit trail of all decisions

6.8 Technology Stack

Table 8: Agentic AI Implementation Stack

Component	Technology
Orchestration	LangGraph (Python)
Data Processing	Pandas, NumPy, Scikit-learn
Visualization	Matplotlib, Seaborn
LLM (Cloud)	Azure OpenAI API (GPT-4.1)
SLM (Local)	HuggingFace Transformers (Qwen2.5-0.5B)
UI Framework	Streamlit
State Persistence	LangGraph MemorySaver (checkpointing)

6.9 Key Features

- **Autonomous Decisions:** Chart type selection and imputation strategy selection based on data characteristics
- **Hybrid AI:** Cloud LLM for reasoning + local SLM for efficiency
- **Explainability:** All decisions logged with justifications
- **Checkpointing:** Fault tolerance via state snapshots
- **Offline Capability:** Local SLM enables operation without internet after first model download
- **Cost-conscious:** Minimal cloud API calls (3 LLM calls per run)
- **Interactive UI:** Real-time Streamlit dashboard showing results

7 Key Insights

7.1 GenAI for Data Synthesis

Finding: GPT-4.1 generated 500 realistic patient records in ≈ 30 minutes. However, post-generation validation was critical (patient ID duplication required correction).

7.2 LLM Vision Analysis

Finding: Azure OpenAI's vision model identified statistical patterns (bimodal age, tachycardia, dengue dominance) but lacked domain context (seasonal epidemiology, digit preference significance).

Implication: AI excels at quantitative pattern extraction; human expertise remains essential for clinical interpretation and epidemiological context.

7.3 Agentic AI Frameworks

Finding: LangGraph pipeline automated chart selection (scatter at $r > 0.6$ vs. boxplot) and imputation strategy selection (mean/ $< 5\%$, KNN/ $5\% - 20\%$, MICE/ $\geq 20\%$).

Implication: Agentic frameworks enable autonomous, explainable workflows with state checkpointing and audit trails. Key advantages: autonomous decisions, context-aware imputation, explainability, and fault tolerance. However, guardrails are essential to prevent inappropriate automated selections.

7.4 SLM vs LLM Performance in Medical Classification

Finding: SLM (Qwen 1.5B) and LLM (GPT-4.1, 170B) achieved only 70% agreement on diagnosis classification. SLM showed systematic bias toward "Chronic Condition" labels (14.0% vs. LLM's 0.2%).

Implication: Substantial performance gap in medical tasks. SLMs offer cost/offline advantages but exhibit domain misclassification (e.g., dengue as chronic), conservative bias, and prompt sensitivity. The 30% disagreement rate requires human validation for critical decisions.

7.5 KNN Imputation Preserves Correlations

Finding: KNN imputation preserved inter-feature correlations (fever $\leftrightarrow$ tachycardia, age $\leftrightarrow$ BP), filling 247 values while maintaining 500 rows.

Implication: ML-based imputation outperforms naive methods in preserving clinical correlation structures.

7.6 Human-AI Complementarity

Finding: Humans excelled at domain context (epidemiology, seasonal patterns, artifacts) while LLMs detected quantitative patterns (digit preference, statistical anomalies).

Implication: Highest-value insights emerge at the human-AI interface. LLMs for pattern extraction; humans for clinical interpretation and context.

7.7 Prompt Engineering and Reproducibility

Finding: Deterministic temperature (0.3 for LLM) stabilized outputs; higher temperatures (0.9 for SLM) increased misclassification.

Implication: Reproducibility requires explicit parameter tuning and version control of prompts. Low temperature (≤ 0.3) recommended for medical tasks to minimize hallucinations.

7.8 Synthetic Data Validation

Finding: GPT-4.1 generated clinically plausible records but exhibited demographic biases (85% female, Malay/Chinese only).

Implication: Synthetic data should augment, not replace, real datasets. Best practice: statistical validation, clinical review, cross-dataset testing, and privacy checks.

8 Conclusion

Generative AI enables modern data pipelines but requires rigorous validation and human oversight. The three-task workflow (synthesis → analysis → preparation) demonstrates practical GenAI integration and opportunities (rapid prototyping, automated insights, agentic orchestration) alongside risks (data quality, SLM limitations, cost scaling).

Key lesson: AI augments, not replaces, data science expertise. Highest value emerges at the human-AI interface. Agentic AI frameworks enable autonomous decision-making while maintaining explainability.

8.1 Key Takeaways

- Synthetic data generation requires rigorous validation; post-hoc QA is critical
- LLMs excel at pattern extraction; humans provide clinical context
- Agentic frameworks enable autonomous, explainable, fault-tolerant pipelines
- SLM/LLM trade-off: cost vs. accuracy (30% disagreement rate observed)
- KNN imputation preserves correlations better than naive methods
- Human-AI interface delivers highest-value insights

References

1. Joel, L.O., Doorsamy, W., & Paul, B.S. (2024). On the Performance of Imputation Techniques for Missing Values on Healthcare Datasets. *arXiv preprint arXiv:2403.14687*. <https://arxiv.org/abs/2403.14687>
2. Lin, K.H., et al. (2025). Diagnostic performance of large language models on the NEJM image challenge: a comparative study with human evaluators. *Frontiers in Medicine*, 12, 1709413. <https://doi.org/10.3389/fmed.2025.1709413>
3. Vassef, S., Goyal, A., Kumar, N., & Saha, K. (2025). Agentic AI framework for End-to-End Medical Data Inference. *arXiv preprint arXiv:2507.18115*. <https://arxiv.org/abs/2507.18115>
4. Lin, K.H., et al. (2025). Benchmarking large language models GPT-4o, llama 3.1, and qwen 2.5 for cancer genetic variant classification. *npj Precision Oncology*, 9(1), 1-12. <https://pubmed.ncbi.nlm.nih.gov/40369023>
5. Stanford HAI. (2025). Holistic Evaluation of Large Language Models for Medical Applications. *Stanford Human-Centered AI Institute*. <https://hai.stanford.edu/news/holistic-evaluation-large-language-models-medical-applications>
6. Razvan, C. & Taran, K. Comparison of the Most Influential Missing Data Imputation Algorithms in Healthcare. *JAIST Technical Report*.
7. Springer Nature. (2025). Review of generative AI for synthetic data creation in healthcare. *Artificial Intelligence Review*, 58(1), 1-45. <https://doi.org/10.1007/s10462-025-11440-2>
8. Zhu, S., Hu, W., Yang, Z., Yan, J., & Zhang, F. (2025). Qwen-2.5 Outperforms Other Large Language Models in the Chinese National Nursing Licensing Examination: Retrospective Cross-Sectional Comparative Study. *JMIR Medical Informatics*, 13, e63731. <https://doi.org/10.2196/63731>

A Code Artifacts

All source code, notebooks, data, and visualizations are available at:

https://github.com/ridhwanrazaliwork/Agentic_medical_AI_pipeline

Key files:

- `assignment1_main.ipynb` – Primary notebook containing all three tasks
- `artifacts/generated_patients.csv` – 500 synthetic patient records
- `artifacts/cleaned_patients_processed.csv` – Post-imputation dataset (0% missing)
- `artifacts/chart_*.png` – Three EDA visualizations
- `prompts/patient_generation_prompt.md` – Patient generation prompt template